higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1
- Oil may be flammable and can cause fire hazards.
- There is a risk of formation of an explosive mixture with the air.
- Due to decomposition of the oil in the arc, the oil becomes polluted by carbon particles.
- Contact erosion (Any 1 × 1) (1)

1.2 A – Boiler/Cooling Tower
B – Turbine
C – Generator
D – Condenser
E – Pump (5 × 1) (5)

1.3 1.3.1 Converts kinetic energy/ Mechanical energy into electrical energy

1.3.2 Uses the heat from burning fuel to make steam and heats steam to drive a steam turbine

1.3.3 Releases exhaust fumes from furnace to air

1.3.4 Transfers heat from a cooling liquid to the air by blowing air over pipes or channels which carry the hot water.

1.3.5 Transfers heat in the steam to the cold water in the tubes. or
Changes the exhaust steam from steam turbine into water.
(5 × 2) (10)

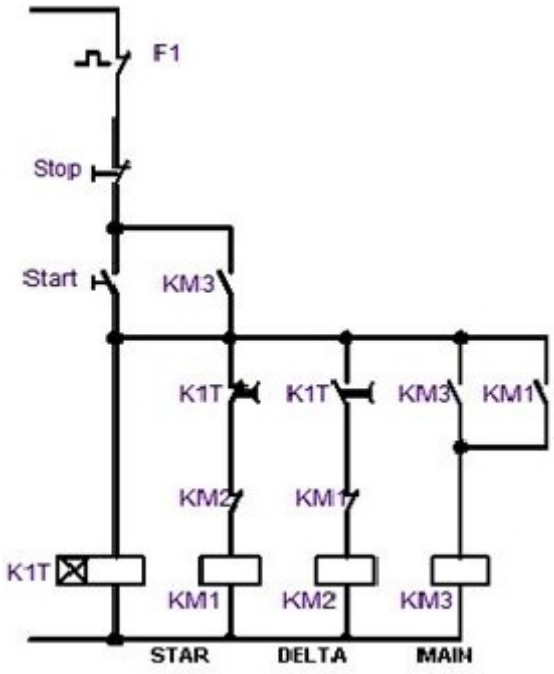
1.4
- Stray insulators
- Composite stray insulators
- Pin insulators
- Composite tension insulators
- Glass tension insulator strings
- Strain insulators.
- Suspension insulators (Any 2 × 1) (2)

1.5
- Rural customers are usually supplied by means of a Radial distribution network.
- It consists of power lines which are fed from one source.
- If there is a power failure it will affect the whole area. (Any 2 × 1) (2)

[20]

QUESTION 2

2.1



K1T, KM1 & KM3 energised
K1T operates
KM3 & KM2 energised

1. Any label=3 marks
2. Correct diagram=3

CM's, IM's and Markers. Please check for similar alternate diagrams.

ADDENDUM: PAGE 7

θ (6)

2.2
- An inductive load uses a magnetic field combined with current.
- A Load which includes a coil (Motor, Solenoid etc.)
- It produces a mechanical force. (Any 2 × 1) (2)

2.3
- Ratings of electrical components
- Conductors and their different paths
- Interconnection of components (Any 2 × 1) (2)

2.4
- Barriers to be erected or the site to be closed off
- Signs need to be put up
- All workers must wear personal protective clothing
- Qualified personnel must be present all the time
- Identify emergency points
- First-Aid kit must be available
- Follow rules prescribed by OHS Act (Any 3 × 1) (3)

2.5
- AC/DC Hipot tests
- Continuity tests/Ground bond
- Insulation resistance tests
- Functional run tests
- Line leakage tests (Any 2 × 1) (2)

[15]

QUESTION 3

3.1
- Voltage
- Current
- Fault effect
- Electromagnetic compatibility (Any 1 × 1) (1)

3.2
- Electrical requirements
- Mechanical requirements
- Environmental requirements (3)

3.3
- It is cheaper
- It has a better resistivity
- It is lighter in weight (3)

3.4 3.4.1 Wind may affect pole supports, stays and clearances.

3.4.2 Lightning may cause voltage surges to the network.

3.4.3 Salt fog can corrode conductors and other system components.
(3 × 2) (6)

3.5
- To determine the foundation type
- To determine pole sizes (2)

[15]

QUESTION 4

4.1
4.1.1 True
4.1.2 False
4.1.3 True
4.1.4 False
4.1.5 True
4.1.6 False
4.1.7 True
4.1.8 False
4.1.9 False
4.1.10 True
(10 × 1) (10)

4.2 4.2.1 A cross-sectional plan shows the building as if it was cut in half.

4.2.2 A floor plan shows the location and type of electrical devices in a building.
(2 × 2) (4)

4.3
- Floor plan
- It shows all electrical devices installed in a building. (2 × 1) (2)

4.4
- Moisture
- Corrosion
- Dirt
- Mechanical damage
- Do they have shrouded connectors?
- Are the test leads intact and insulated properly?
- Are they double insulated
- Do they has as little exposed metal at the tips as possible?
- Are their CAT ratings the same or higher than the meter rating? (Any 2 × 1) (2)

4.5
- It will burn the skin.
- It might cause the person to be shocked. (Any 2 × 1) (2)

[20]

QUESTION 5

5.1
- Fault circuit indicators are clamped around a cable to measure the current.
- It indicates where the faulty current is. (2 × 1) (2)

5.2
- High currents
- Short circuits
- Humidity
- Mechanical damage (Any 2 × 1) (2)

5.3
- Excess current in the neutral of the transformer
- Problems in the cooling system
- High harmonic content in the power supply
- Sustained overvoltage
- Grounded windings
- A lack of oil in the transformers
- Breakdown in the insulating material (Any 3 × 1) (3)

5.4

CAUSE	REMEDY
Low voltage	Check coil terminal voltage
Poor contact in control circuit	Replace the device

(4 × 1) (4)

5.5
- Clean the motor
- Check for heating
- Check bearing condition
- Check for accurate shaft alignment and keyway
- Check for loose screws
- Check for broken bases (Any 4 × 1) (4)

[15]

QUESTION 6

6.1
- It is a group of modules or panels wired together
- to produce the desired voltage. (2 × 1) (2)

6.2 6.2.1

$$Power-consumption_{BULBS} = (8x8) \times 5 / 1000$$
$$Power-consumption_{BULBS} = 0{,}32\ kWh/day$$ ✓✓

$$Power-consumption_{M/WAVE} = 1\,300 \times 1{,}0 / 1\,0($$ ✓✓
$$Power-consumption_{M/WAVE} = 1{,}30\ kWh/day$$

$$Power-consumption_{GEYSER} = 1\,500 \times 3 / 1\,000$$
$$Power-consumption_{LIGHTS} = 4{,}5\ kWh/day$$ ✓✓ (6)

6.2.2
$$Power-consumption_{TOTAL} = 0{,}32 + 1{,}3 + 4{,}5$$
$$Power-consumption_{TOTAL} = 6{,}12\ kWh/day$$ ✓✓ (2)

6.2.3
$$Amount-To-Be-Saved_{COST} = 6{,}12 \times 30 \times 2{,}05$$
$$Amount-To-Be-Saved_{COST} = R376{,}38/mont$$ ✓✓ (2)

6.3
- The percentage of the capacity of the batteries
- to which they can be discharged without being damaged (2 × 1) (2)

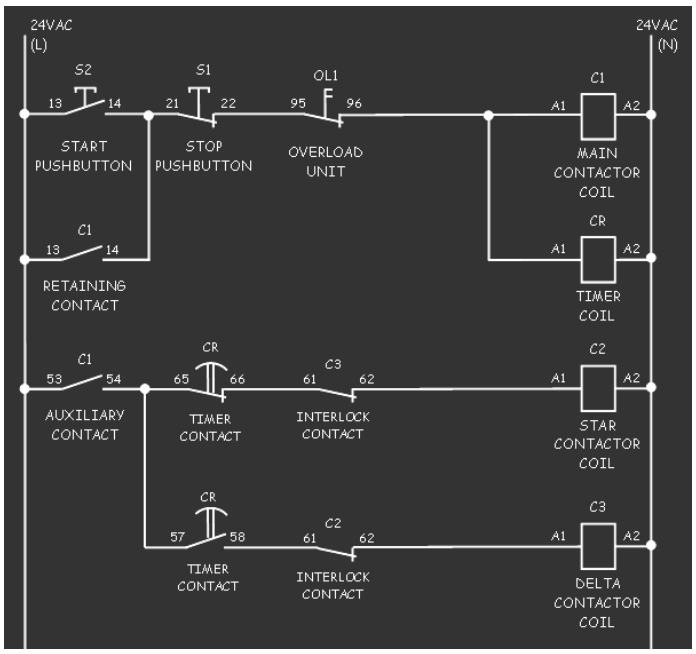
6.4 To increase voltage or amperage or both voltage and amperage (1)

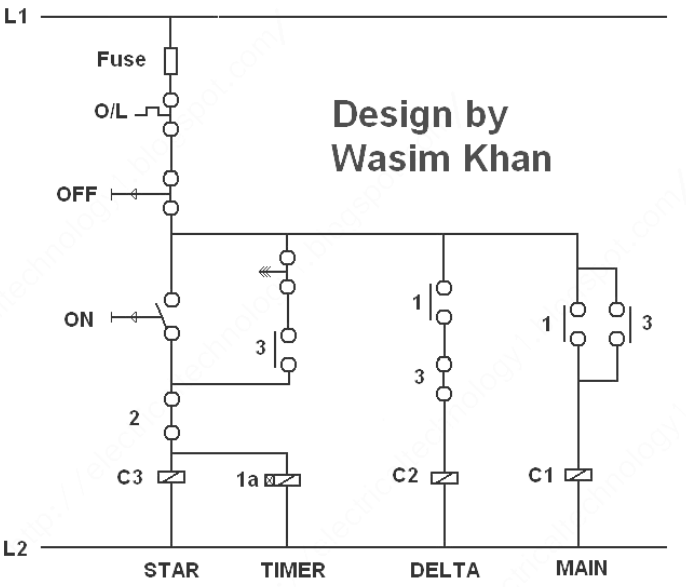
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TOTAL: 100

ADDENDUM

QUESTION 2.1





Control Circuit Diagram of Automatic Star Delta (Y-Δ) starter with Timer for 3-Phase Motor.

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